

Target Hamiltonian

ω_q, α

ω_r, g, κ

Physics targets

Physics features

$$\omega_q \approx \sqrt{8E_J E_C} - E_C$$
$$\alpha \approx -E_C$$

Scale and encode

Min-max scaling

Categorical masks

ML

Inverse MLPs

TransmonCross

Ncap Coupler

Resonator

Iterate

Validation

Compare

Percent error

Iterate if needed

Recovered Hamiltonian

$\hat{\omega}_q, \hat{\alpha}$

$\hat{\omega}_r, \hat{g}$

Forward check

Quantum Metal layout

Ansys or surrogate

Decode predictions

Physical units

Valid design fields